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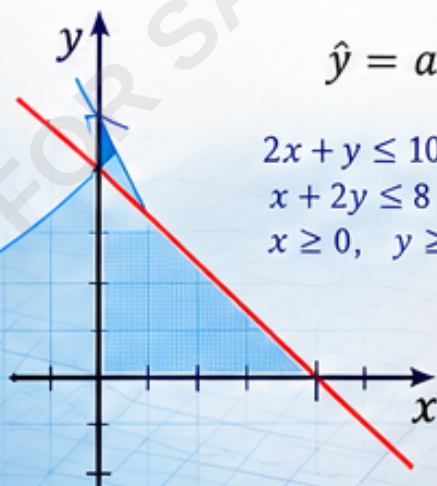
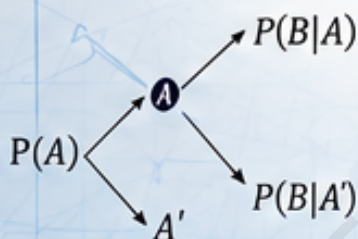
Core Mathematics

Book 1

Concepts, Notes & Application

Clear Notes | Worked Examples | Topical Practice
Mixed Concepts | 6 Complete Practice Tests

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & -2 \end{pmatrix}$$



$$\hat{y} = a + bx$$

$$\begin{aligned} 2x + y &\leq 10 \\ x + 2y &\leq 8 \\ x &\geq 0, \quad y &\geq 0 \end{aligned}$$



$$\sin^2 \theta + \cos^2 \theta = 1$$



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PEAK ACHIEVERS ONLINE ACADEMY

GRADE 12 CORE MATHEMATICS

REVISION BOOK

BOOK 1 - CONCEPTS, NOTES & APPLICATION

Concise CBE-oriented notes • Worked examples • Applications • Mixed Concepts • Six 100-mark practice tests

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CHAPTER 1: MATRICES II

A matrix is a rectangular arrangement of numbers. Its order is rows $\times$ columns.

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\det(A) = ad - bc$. A matrix is singular when $\det(A) = 0$.

For a non-singular 2×2 matrix, $A^{-1} = \frac{1}{(ad-bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Simultaneous equations can be written $AX=B$ and solved using $X=A^{-1}B$.

REMEMBER: Show the rule or formula, substitute carefully, and state the final answer with units where required.

Worked Examples

Determinant

For $A = \begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix}$, $\det(A) = 5(4) - 3(2) = 14$.

Inverse

For $A = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$, $\det(A) = 14$, so $A^{-1} = \frac{1}{14} \begin{bmatrix} 5 & -3 \\ -2 & 4 \end{bmatrix}$.

Simultaneous equations

$2x + 3y = 13$ and $x + 2y = 8$ give $x = 2$, $y = 3$.

Application

2 books + 3 pens = KSh210 and 3 books + 2 pens = KSh240. Solving gives book = KSh60, pen = KSh30.

Topical Check

Attempt at least one routine item and one application item from this chapter before moving on.

THINK BIG: In an examination, method selection and interpretation are assessed alongside arithmetic accuracy.

DEEPEENED CONCEPT DEVELOPMENT

Learning focus: Use matrices confidently as a method for organising information and solving two-variable problems.

Matrix multiplication

If A is $m \times n$ and B is $n \times p$, then AB exists and has order $m \times p$. Multiply each row of A by each column of B .

Identity matrix

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. For a compatible matrix A , $AI = IA = A$. For a non-singular matrix, $AA^{-1} = A^{-1}A = I$.

Why the determinant matters

For a 2×2 matrix, $\det(A) = ad - bc$. If $\det(A) = 0$, the rows/columns are dependent and a unique inverse does not exist.

HINT: Write the governing rule before substituting numbers. This makes method marks visible.

REMEMBER: A correct final answer without supporting working may lose method marks in a structured question.

CAUTION: Check signs, units, domain restrictions and rounding only at the final stage.

SPOTLIGHT: Connect the mathematics to the situation: explain what the number means, not only what it equals.

More Worked Examples

Matrix product

$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 \\ 2 & -1 \end{bmatrix}$. $AB = \begin{bmatrix} 2(1)+1(2) & 2(5)+1(-1) \\ 3(1)+4(2) & 3(5)+4(-1) \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 11 & 11 \end{bmatrix}$.

Inverse check

$A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$, $\det = 6 - 2 = 4$. $A^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -1 \\ -2 & 3 \end{bmatrix}$. Multiplying AA^{-1} gives I , confirming the inverse.

CBE transport application

Two vans carry adults and learners. $2a + 3l = 96$ and $3a + 2l = 104$. Matrix method gives $a = 24$ and $l = 16$. Interpretation: the unit group values are 24 adults and 16 learners.

Topical Practice and Fully Worked Answers

1. Find $\det(\begin{bmatrix} 7 & 2 \\ 3 & 5 \end{bmatrix})$.

Solution: $7(5) - 2(3) = 35 - 6 = 29$.

2. Find the inverse of $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$.

Solution: $\det = 6 - 1 = 5$; inverse $= \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$.

3. Multiply $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ by $\begin{bmatrix} 4 & 1 \\ 2 & 5 \end{bmatrix}$.

Solution: $\begin{bmatrix} 1(4)+2(2) & 1(1)+2(5) \\ 0(4)+3(2) & 0(1)+3(5) \end{bmatrix} = \begin{bmatrix} 8 & 11 \\ 6 & 15 \end{bmatrix}$.

4. Solve $2x + y = 11$, $x + 3y = 13$.

Solution: From first $y = 11 - 2x$. Substitute: $x + 3(11 - 2x) = 13 \rightarrow -5x = -20 \rightarrow x = 4$, $y = 3$.

5. State why $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ has no inverse.

Solution: $\det = 2(2) - 4(1) = 0$, so it is singular.

6. A shop sells 3 rulers+2 sets for KSh460 and 2 rulers+3 sets for KSh540. Find prices.

Solution: $3r+2s=460$; $2r+3s=540$. Elimination gives $5r=300 \rightarrow r=60$; then $s=140$.

THINK BIG: Change one value in a worked example and solve it again. This tests whether you understand the method rather than memorising the answer.

CHAPTER 2: VECTORS II

A vector has magnitude and direction. $AB = OB - OA$.

For $v=(a,b)$, $|v|=\sqrt{a^2+b^2}$.

If $AP:PB=m:n$, then $OP=(n OA + m OB)/(m+n)$.

Parallel vectors are scalar multiples; collinearity can be proved using proportional direction vectors.

REMEMBER: Show the rule or formula, substitute carefully, and state the final answer with units where required.

Worked Examples

Internal division

$A=(2,3)$, $B=(8,12)$, $AP:PB=2:1$ gives $P=(6,9)$.

Displacement

8 km east and 6 km north gives magnitude $\sqrt{8^2+6^2}=10$ km and direction $\tan^{-1}(6/8)=36.9^\circ$ north of east.

Topical Check

Attempt at least one routine item and one application item from this chapter before moving on.

THINK BIG: In an examination, method selection and interpretation are assessed alongside arithmetic accuracy.

DEEPENED CONCEPT DEVELOPMENT

Learning focus: Represent displacement, division of line segments and geometric relationships using vectors.

Position and displacement

If $OA=a$ and $OB=b$, then $AB=b-a$. Direction matters: $BA=a-b=-AB$.

Scalar multiples

Vectors u and v are parallel when $u=kv$ for some scalar k . If $k<0$ they point in opposite directions.

Section formula

If P divides AB internally in ratio $AP:PB=m:n$, then $OP=(n a+m b)/(m+n)$.

HINT: Write the governing rule before substituting numbers. This makes method marks visible.

REMEMBER: A correct final answer without supporting working may lose method marks in a structured question.

CAUTION: Check signs, units, domain restrictions and rounding only at the final stage.

SPOTLIGHT: Connect the mathematics to the situation: explain what the number means, not only what it equals.

More Worked Examples

Collinearity

$A(1,2)$, $B(4,8)$, $C(7,14)$. $AB=(3,6)$, $BC=(3,6)$, so the points are collinear.

Internal division

$A=(-2,4)$, $B=(8,9)$, $AP:PB=3:2$. $P=[2A+3B]/5=[(-4,8)+(24,27)]/5=(4,7)$.

Navigation

A boat moves 12 km east then 5 km south. Resultant= $(12,-5)$, magnitude=13 km, direction $\tan^{-1}(5/12)=22.6^\circ$ south of east.

Topical Practice and Fully Worked Answers

1. Find AB for $A(-3,2)$, $B(5,8)$.

Solution: $AB=(8,6)$.

2. Find $| (9,12) |$.

Solution: $\sqrt{81+144}=15$.

3. Are $(6,9)$ and $(2,3)$ parallel?

Solution: Yes. $(6,9)=3(2,3)$.

4. Find midpoint of $A(2,-1)$, $B(10,7)$.

Solution: $((2+10)/2, (-1+7)/2)=(6,3)$.

5. P divides $A(1,2)$, $B(11,7)$ in ratio $2:3$. Find P .

Solution: $P=[3A+2B]/5=[(3,6)+(22,14)]/5=(5,4)$.

6. A hiker walks 7 km north and 24 km east. Find resultant.

DEEPENED CONCEPT DEVELOPMENT

Learning focus: Describe association, calculate regression models and make cautious predictions.

Correlation

r close to 1: strong positive; close to -1: strong negative; close to 0: weak linear association.

Regression

The line $y=a+bx$ predicts y from x ; do not reverse roles without recalculating.

Prediction limits

Interpolation is generally safer than extrapolation because it stays within observed data.

HINT: Write the governing rule before substituting numbers. This makes method marks visible.

REMEMBER: A correct final answer without supporting working may lose method marks in a structured question.

CAUTION: Check signs, units, domain restrictions and rounding only at the final stage.

SPOTLIGHT: Connect the mathematics to the situation: explain what the number means, not only what it equals.

More Worked Examples

Regression from summaries

$n=5, \Sigma x=20, \Sigma y=35, \Sigma x^2=90, \Sigma xy=160$. $b=(800-700)/(450-400)=2$; $a=7-2(4)=-1$. $y=2x-1$.

Prediction

Using $y=2x-1$, at $x=6$, $y=11$.

Interpretation

$r=-0.91$ indicates strong negative linear association, but it does not establish that x causes y .

Topical Practice and Fully Worked Answers

1. If $r=.95$, describe.

Solution: Strong positive linear correlation.

2. If $r=-.12$.

Solution: Very weak negative linear correlation.

3. $y=3+4x$ at $x=5$.

Solution: 23.

4. $\bar{x}=4, \bar{y}=10, b=1.5$. Find a .

Solution: $10-1.5(4)=4$.

5. Regression line.

Solution: $y=4+1.5x$.

6. Why caution at x far outside sample?

Solution: It is extrapolation; the observed relationship may not continue.

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THINK BIG: Change one value in a worked example and solve it again. This tests whether you understand the method rather than memorising the answer.

CHAPTER 14: NUMERICAL METHODS

Numerical methods approximate roots that may be difficult to obtain algebraically.

Bisection requires a continuous function with opposite signs at the ends of an interval.

At each step, evaluate the midpoint and retain the half-interval containing the sign change.

Continue until the required interval width or decimal accuracy is achieved.

REMEMBER: Show the rule or formula, substitute carefully, and state the final answer with units where required.

Worked Examples

Bisection

For $f(x)=x^3-2x-5$, $f(2)=-1$ and $f(3)=16$, so a root lies in $(2,3)$. At 2.5, $f=5.625$, so the root lies in $(2,2.5)$.

Topical Check

Attempt at least one routine item and one application item from this chapter before moving on.

THINK BIG: In an examination, method selection and interpretation are assessed alongside arithmetic accuracy.

CHAPTER 16: MIXED CONCEPTS II - CALCULUS, STATISTICS & PROBABILITY

These questions deliberately mix concepts. Identify the method before calculating.

1. Differentiate $y=x^3-6x^2+9x+4$ and find stationary x -values.
2. Evaluate $\int_0^3(2x^2+4x)dx$.
3. For 5,7,9,11,13 find the population variance and standard deviation.
4. A bag contains 5 red and 3 blue balls. Find $P(\text{two red without replacement})$.
5. A rectangular field has perimeter 120 m. Use differentiation to find its maximum area.

Answer Guide

1. $y'=3(x-1)(x-3)$; stationary $x=1,3$.
2. 36.
3. variance=8; standard deviation ≈ 2.83 .
4. $5/8 \times 4/7 = 5/14$.
5. 30 m $\times$ 30 m; maximum area=900 m².

AUDITED COMPLETE WORKED SOLUTIONS - TESTS 1 TO 3

TEST 1 - Complete Step-by-Step Marking Scheme

Question 1

$\det = 5(4) - 2(3) = 14$. Therefore $A^{-1} = (1/14)[[4, -2], [-3, 5]]$.

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 2

$2a - b = 2(3, -2) - (5, 4) = (6, -4) - (5, 4) = (1, -8)$. Magnitude $= \sqrt{1 + 64} = \sqrt{65} \approx 8.06$.

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 3

Midpoint of $(2, 4), (8, 4)$ is $(5, 4)$. AB is horizontal, so its perpendicular bisector is vertical: $x = 5$.

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 4

$[[0, -1], [1, 0]](4, -3)^T = (3, 4)^T$. Hence image $= (3, 4)$, a 90° anticlockwise rotation.

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 5

$P(A \cup B) = 0.65 + 0.50 - 0.30 = 0.85$.

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 6

$y' = 8x^3 - 15x^2 + 3$. At $x = 2$: $64 - 60 + 3 = 7$.

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 7

$\int_{-1}^3 (3x^2 + 2x) dx = [x^3 + x^2]_{-1}^3 = (27 + 9) - (1 + 1) = 34$.

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 8

$$A = 80000(1.075)^3 = 80000(1.242296875) = \text{KSh}99,383.75.$$

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 9

$$2c^2 - c - 1 = 0 = (2c + 1)(c - 1). \text{ Thus } \cos\theta = 1 \text{ or } -1/2, \text{ giving } \theta = 0^\circ, 120^\circ, 240^\circ, 360^\circ.$$

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 10

$$f(2) = -1, f(3) = 15. \text{ Midpoint } 2.5 \text{ gives } f(2.5) = 5.125 > 0, \text{ so root is in } (2, 2.5).$$

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 11

Constraints $x + y \leq 30, 2x + y \leq 40$. Intersection = (10, 20). Vertices (0, 0), (20, 0), (10, 20), (0, 30). $P = 600x + 900y$ gives 0, 12000, 24000, 27000. Max KSh27,000 at (0, 30).

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 12

$$\text{Floor diagonal} = \sqrt{18^2 + 12^2} = \sqrt{468} \approx 21.63\text{m}. \text{ Space diagonal} = \sqrt{468 + 8^2} = \sqrt{532} \approx 23.07\text{m}. \tan\theta = 8/21.63, \theta \approx 20.3^\circ.$$

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 13

Both means = 50. A squared deviations sum = 10, variance = 2, $sd \approx 1.41$. B squared deviations sum = 90, variance = 18, $sd \approx 4.24$. A is more consistent.

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 14

$$2x + 2y = 120 \rightarrow y = 60 - x. A = 60x - x^2. A' = 60 - 2x = 0 \rightarrow x = 30, y = 30. A'' = -2 < 0. \text{ Max area} = 900\text{m}^2.$$

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.

Question 15

Simple: $250000[1 + 0.10(4)] = 350000$. Compound: $250000(1.09)^4 = 352895.40$. Compound exceeds simple by KSh2,895.40.

Marking: award method marks for the correct formula/setup and intermediate working, then the accuracy/final-answer mark.



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The *Peak Achievers* Series is a focused set of revision and study books designed to help learners master key concepts, develop problem-solving skills and excel in their examinations. Each book is written in a clear, structured and student-friendly style with worked examples, topical practice, real-life applications and examination-style questions.

ABOUT THIS BOOK

Grade 12 Core Mathematics Book 1: Concepts, Notes & Application

is the first book in the Grade 12 Mathematics series. It develops core concepts in a clear and progressive manner, with fully worked examples, helpful hints, important notes, common errors, real-life applications and topical practice. The book also includes Mixed Concepts and six full National Examination–Style Practice Tests with complete, worked marking schemes.

KEY FEATURES

- ✓ Clear and concise explanatory notes
- ✓ Step-by-step worked examples
- ✓ HINT, REMEMBER, CAUTION, SPOTLIGHT and THINK BIG boxes
- ✓ Real-life and CBE-style applications
- ✓ Topical practice with fully worked answers
- ✓ Mixed Concepts for deeper understanding
- ✓ Six 100-mark National Examination–Style Practice Tests with worked marking schemes
- ✓ Written by an experienced learner and verified by a qualified Mathematics teacher

CHAPTERS IN BOOK 1

- | | |
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| 1. Matrices II | 8. Differentiation II |
| 2. Vectors II | 9. Integration I |
| 3. Loci | 10. Trigonometry III |
| 4. Linear Programming | 11. Three-Dimensional Geometry |
| 5. Transformations III | 12. Financial Mathematics |
| 6. Measures of Dispersion | 13. Correlation and Regression |
| 7. Probability III | 14. Revision and Application |

Mixed Concepts I – IV | Practice Tests 1 – 6 (100 marks each) with full solutions

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